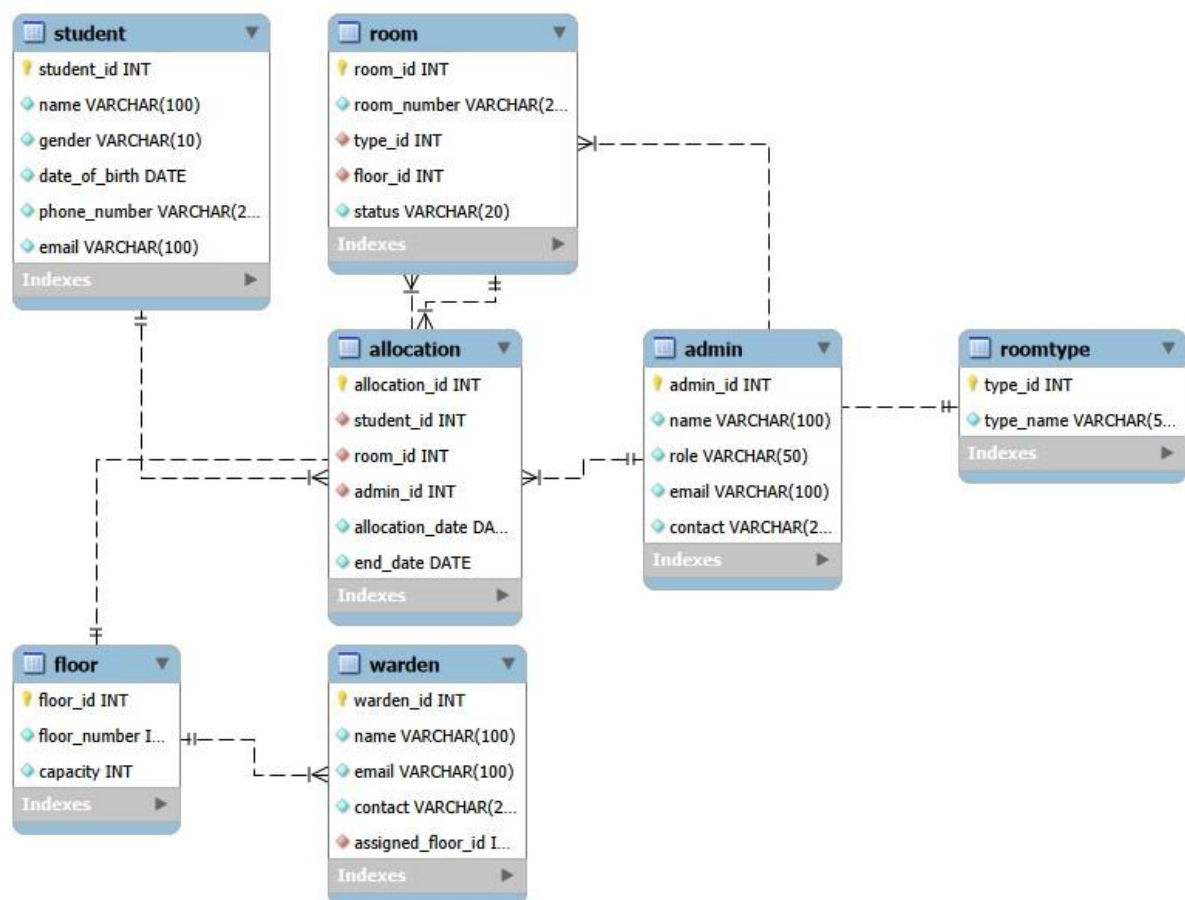


Group Information

- **Group Name:** HostelRoomAllocation_Group01
 - **Members:** Kashaf Sher & Alishba Sadeeq
 - **Project:** Hostel Room Management System
 - **Course:** Database Systems Lab
 - **Date:** 17th May 2026
-

Updated ERD Diagram



- Entities: Admin, Warden, Floor, Room, RoomType, Student, Allocation.
- Show PKs, FKs, and relationships.
- Note: Duplicate Room entity merged, Student contact split into phone/email.

Normalization Walkthrough (1NF → 3NF)

Summarize from Milestone 2:

- **1NF:**

- Removed duplicate Room entity.
 - Split Student.contact_info into phone_number + email.
- **2NF:** ○ All tables have single-column PKs → no partial dependencies.
- **3NF:**
 - Checked for transitive dependencies.
 - Room.status retained as justified denormalization for performance.
- **Outcome:** Schema fully normalized to 3NF.

Dataflow Description (from Milestone 3)

- **Actors:** Admins, Wardens, Students, System.
- **Flow:** ○ RoomType → Floor → Room → [Admin + Student] → Allocation.
- **Process:**
 - Admin sets up rooms and wardens. ○ Students register. ○ Admin allocates rooms → Room status updated.
 - Wardens query occupancy.
- **Outputs:** Availability reports, allocation reports, floor summaries

Database Setup (DDL) (from milestone 4)

- Created 7 tables (RoomType, Floor, Admin, Warden, Room, Student, Allocation).
- Defined PKs, FKs, UNIQUE, CHECK, NOT NULL constraints.
- Verified schema in MySQL Workbench via EER diagram.
- Fixed duplicate index errors by removing redundant CREATE INDEX lines.

Data Population (DML) (from milestone 5)

- Populated tables using INSERT statements with synthetic data.
- Demonstrated UPDATE (Room.status → 'Occupied') and DELETE (Student record) .
- Verified row counts and NULL integrity using COUNT(*) and WHERE IS NULL queries.
- Confirmed foreign key consistency via JOIN queries between Allocation, Student, Room, and Admin.

GitHub Repository Link <https://github.com/kashafsher/HostelRoomAllocationSystem.git>

Milestone	Focus	Key Deliverables	Outcome
2	Normalization	ERD refined to 3NF	Clean schema
3	Data Preprocessing	Synthetic dataset + CSV exports	Ready for loading
4	DDL Setup	CREATE TABLE scripts + constraints	Database structure verified
5	DML Population	INSERT/UPDATE/DELETE + JOIN validation	Data loaded successfully
